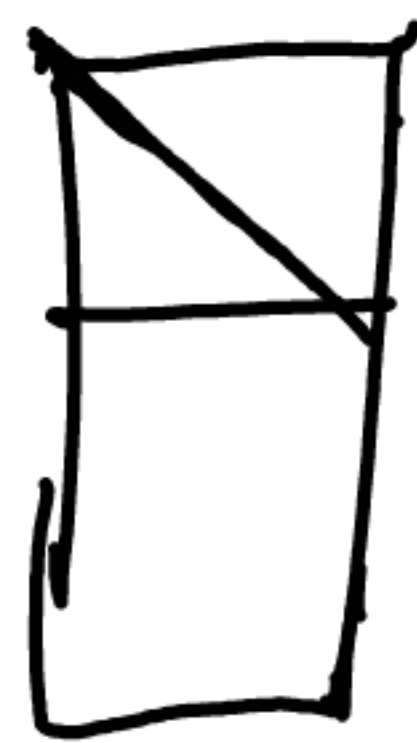


$$A \in \mathbb{R}^{m \times n}, \quad m > n$$

$$A = QR, \quad Q \in \mathbb{R}^{m \times m}, \quad R \in \mathbb{R}^{m \times n}$$

$R$  es triangular superior

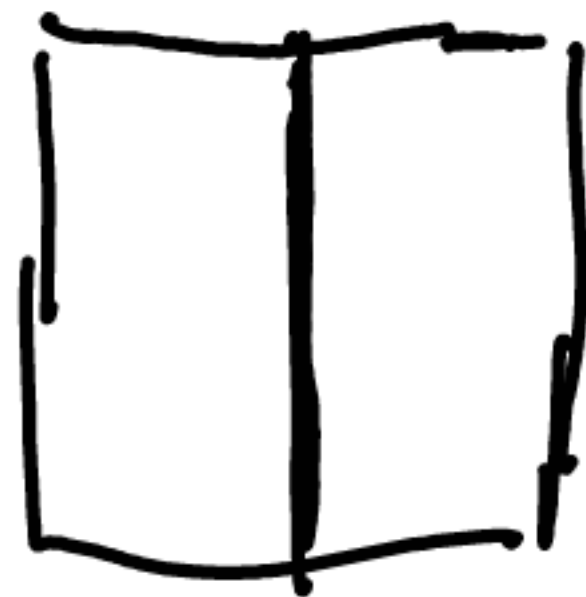


$R$

$$= \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

$$R_1 \in \mathbb{R}^{n \times n}$$

$$0 \in \mathbb{R}^{(m-n) \times n}$$



$Q$

$$= [Q_1 | Q_2]$$

$$Q_1 \in \mathbb{R}^{m \times n}$$

$$Q_2 \in \mathbb{R}^{m \times (m-n)}$$

$$QR = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_1 \\ 0 \end{bmatrix} = \begin{bmatrix} Q_1 R_1 & \cancel{Q_2 0} \end{bmatrix} = A$$

$Q_1 R_1$  forma reduzida de  $QR$ .

## Quadrados Mínimos

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2 = \min_{x \in \mathbb{R}^n} \|Q^T Ax - Q^T b\|_2$$

# Cuadrados Mínimos

$$\min \|Ax - b\|_2$$

↙  
Ecuaciones Normales

$$\boxed{A^T A} x = A^T b$$

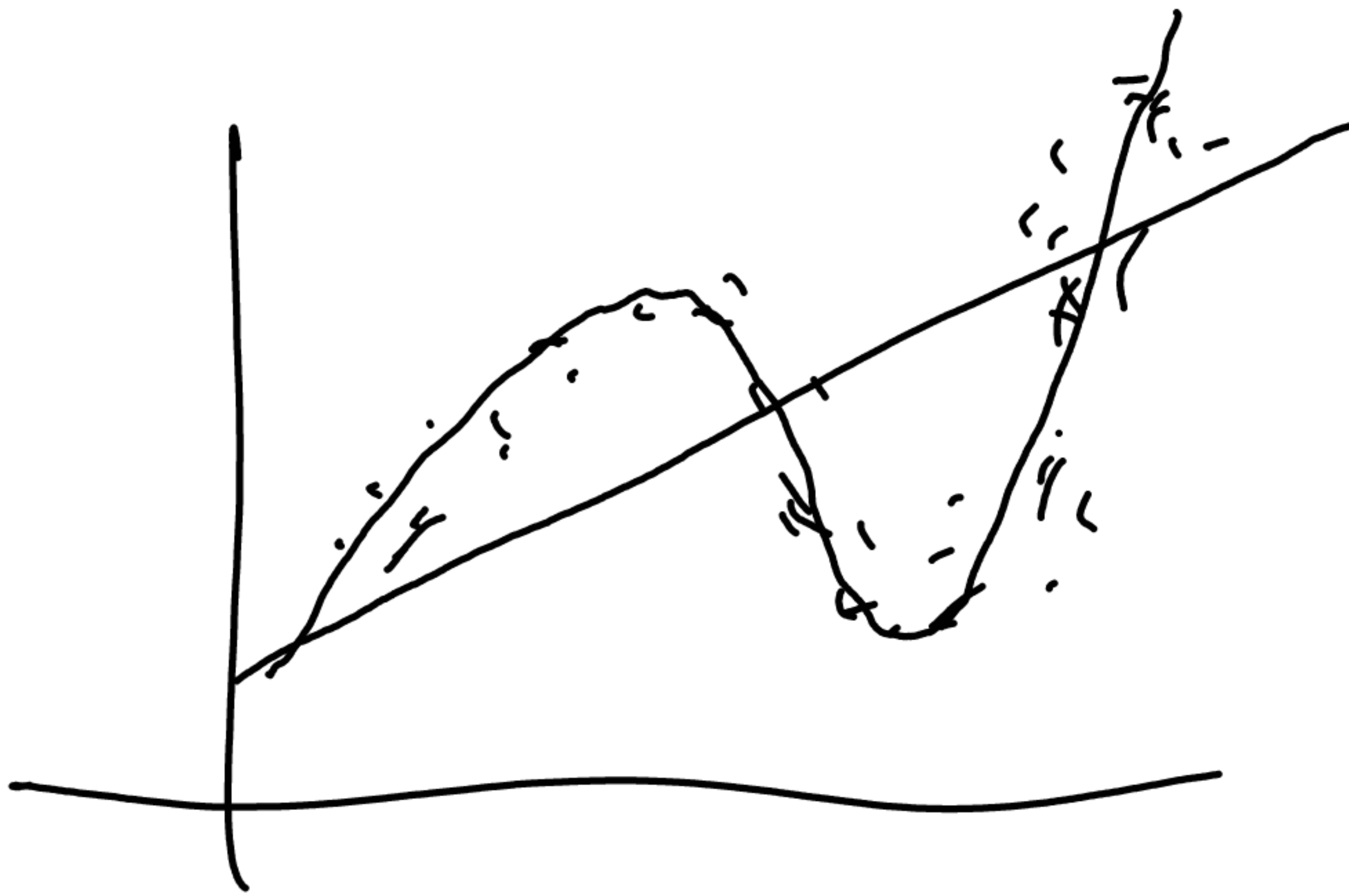
↑  
Cholesky

$\kappa_2$  se eleva al cuadrado

↘  
QR

$$\|Ax - b\|_2 = \|Qx - c\|_2$$

$$\kappa_2(R_1) = \kappa_2(A)$$



## Regresión Polinómica

Aproximar mi nube de puntos con la familia  
de funciones  $\{1, x, x^2, \dots, x^m\}$

$$(t_i, b_i) \quad A = \begin{bmatrix} 1 & t_1 & t_1^2 & \dots & t_1^k \\ 1 & t_2 & t_2^2 & \dots & t_2^k \\ \vdots & & & & \\ 1 & t_m & t_m^2 & \dots & t_m^k \end{bmatrix}$$

Vandermonde